Chengsong Huang

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EDUCATION

Washington University in St. Louis — Sep. 2023 - Present
Advisor: Prof. Jiaxin Huang — St. Louis, MO
Ph.D. in Computer Science (GPA 4.0/4.0)

Fudan University — Sep. 2019 - Jun. 2023
Software School of Fudan University — Shanghai, China
B.S. in Software Engineering (GPA 3.82/4.0) rank 1%

RESEARCH INTERESTS

My research focuses on advancing the capabilities and accessibility of Large Language Models (LLMs). My interests include:

- **Data-Driven Self-Evolution:** Designing frameworks that enable LLMs to evolve themselves through the data and feedback they generate. This includes self-evolving training loops that synthesize their own training data and curricula from zero human-labeled data, self-rewarding models that turn their own reasoning into supervision, and environments that co-evolve with the agents trained in them. Representative works: R-Zero (ICLR 2026), a reasoning LLM that self-evolves from zero data, and EnvHarness (Google Research), which awakens static environments for agent learning.
- **Efficient LLMs:** Developing and analyzing methods to make LLMs more computationally efficient. My experience includes work on parameter-efficient fine-tuning (PEFT), speculative decoding, and adaptive test-time scaling to accelerate training and inference.

PREPRINTS (SELECTED)

** stands for equal contribution or alphabetical ordering.*

Data-Driven Self-Evolution

- **Huang, C.**, Wang, Z., Han, R., Yan, J., Chen, Y., CuiZhu, Z., Jiang, K., Xia, P., Yu, H., Zhuang, Y., Ming, Y., Pan, J., Dalvi Mishra, B., Huang, J., Gokturk, B., Pfister, T. and Lee, C.-Y. "EnvHarness: Awakening Static Worlds for Agent Learning". arXiv preprint. 2026.
- **Huang, C.**, Liu, H., Zheng, T., Dai, R., Huang, L., Li, J., Li, Z., Wei, Z., Meng, Y. and Huang, J. "G-Zero: Self-Play for Open-Ended Generation from Zero Data". arXiv preprint. 2026.
- Zheng, T., Liu, H., **Huang, C.**, Bao, H., Zhang, S., Liu, R., Dai, R., Chen, R., Liu, C., Xiong, T., Wu, X., Zhang, H. and Huang, H. "LLMs Improving LLMs: Agentic Discovery for Test-Time Scaling". arXiv preprint. 2026.
- Yu, W., Liang, Z., **Huang, C.**, Panaganti, K., Fang, T., Mi, H. and Yu, D. "Guided Self-Evolving LLMs with Minimal Human Supervision". arXiv preprint. 2025.

Others

- Li, J., Huang, L., **Huang, C.**, Xu, S., Cai, D., Yang, Y., Zhang, W. and Huang, J. "Process Rewards with Learned Reliability". arXiv preprint. 2026.
- Qian, Q.*, **Huang, C.***, Xu, J., Lv, C., Wu, M., Liu, W., Wang, X., Wang, Z., Huang, Z., Tian, M., Xu, J., Hu, K., Wang, H.-D., Hu, Y., Huang, X. and Zheng, X. "Benchmark^2: Systematic Evaluation of LLM Benchmarks". arXiv preprint. 2026.
- Tang, Y., **Huang, C.**, Huang, J. and Yeoh, W. "UniRel-R1: RL-tuned LLM Reasoning for Knowledge Graph Relational Question Answering". arXiv preprint. 2025.
- Gu, C., **Huang, C.**, Zheng, X., Chang, K.-W. and Hsieh, C.-J. "Watermarking Pre-trained Language Models with Backdooring". arXiv preprint. 2022.

PUBLISHED PAPERS (SELECTED)

For a complete list of publications, please visit my Google Scholar profile.
** stands for equal contribution or alphabetical ordering.*

Data-Driven Self-Evolution

- **Huang, C.**, Yu, W., Wang, X., Zhang, H., Li, Z., Li, R., Huang, J., Mi, H. and Yu, D. "R-Zero: Self-Evolving Reasoning LLM from Zero Data". ICLR 2026.
- He, Y.*, **Huang, C.***, Li, Z.*, Huang, J. and Yang, Y. "VisPlay: Self-Evolving Vision-Language Models from Images". CVPR 2026.
- Li, Z., Du, H., **Huang, C.**, Yu, L., Liu, Z., Wu, X., He, Y., Xie, J., Wu, X., Zhang, J. and Liu, F. "Self-Evolving Vision Language Models for Structured Visual Understanding". EMNLP 2026.

Efficient LLMs

- Zheng, T.*, **Huang, C.***, Dai, R.*, He, Y., Liu, R., Ni, X., Bao, H., Wang, K., Zhu, H., Huang, J., Huang, F. and Huang, H. "Parallel-Probe: Towards Efficient Parallel Thinking via 2D Probing". ICML 2026.
- Li, J., **Huang, C.**, Huang, L., Xu, S., Liu, H., Zhang, W. and Huang, J. "Training Data Efficiency in Multimodal Process Reward Models". ICML 2026.
- **Huang, C.**, Huang, L., Leng, J., Liu, J. and Huang, J. "Efficient Test-Time Scaling via Self-Calibration". ICLR 2026.
- **Huang, C.**, Zheng, T., Huang, L., Li, J., Liu, H. and Huang, J. "RelayLLM: Efficient Reasoning via Collaborative Decoding". Findings of EMNLP 2026.
- **Huang, C.**, Huang, L. and Huang, J. "Divide, Reweigh, and Conquer: a Logit Arithmetic Approach for In-Context Learning". EACL 2026.
- Huang, L., **Huang, C.**, Leng, J., Huang, D. and Huang, J. "POSS: Position Specialist Generates Better Draft for Speculative Decoding". COLM 2026.
- **Huang, C.***, Liu, Q.*, Lin, B. Y.*, Pang, T., Du, C. and Lin, M. "LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition". COLM 2024.

Others

- Li, Z., Yu, W., **Huang, C.**, Liang, Z., Liu, R., Liu, F., Che, J., Yu, D., Boyd-Graber, J., Mi, H. and Yu, D. "Self-Rewarding Vision-Language Model via Reasoning Decomposition". ICLR 2026.
- Huang, L., **Huang, C.**, Li, J., Cai, D., Yang, Y. and Huang, J. "Nonsense Helps: Prompt Space Perturbation Broadens Reasoning Exploration". Findings of EMNLP 2026.
- Leng, J., **Huang, C.**, Huang, L., Lin, B. Y., Cohen, W. W., Wang, H. and Huang, J. "CrossWordBench: Evaluating the Reasoning Capabilities of LLMs and LVLMs with Controllable Puzzle Generation". COLM 2025.
- Leng, J., **Huang, C.**, Zhu, B. and Huang, J. "Taming Overconfidence in LLMs: Reward Calibration in RLHF". ICLR 2025.
- Chen, L., **Huang, C.**, Zheng, X., Lin, J. and Huang, X. "TableVLM: Multi-modal Pre-training for Table Structure Recognition". ACL 2023.
- Lin, B. Y.*, **Huang, C.***, Liu, Q., Gu, W., Sommerer, S. and Ren, X. "On Grounded Planning for Embodied Tasks with Language Models". AAAI 2023.

EXPERIENCE

Google Cloud AI — Feb. 2026 - present
Student Researcher (Mentor: Zifeng Wang)
EnvHarness: Awakening Static Worlds for Agent Learning

Tencent AI Lab Seattle — Jun. 2025 - Aug. 2025
Research intern (Mentor: Wenhao Yu)
R-Zero: Self-Evolving Reasoning LLM from Zero Data

Sea AI lab (SAIL) — Jan. 2023 - Jun. 2023
Research intern (Mentor: Qian Liu)
LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition

USC INK lab — May. 2022 - Jan. 2023
Research intern (Mentor: Xiang Ren, Bill Yuchen Lin)

On Grounded Planning for Embodied Tasks with Language Models

OPEN-SOURCE IMPACT

- Author and maintainer of open-source research code with **2,100+ GitHub stars** in total, including R-Zero (800+ stars), LoraHub (650+ stars), EnvHarness (500+ stars), VisPlay, RelayLLM, G-Zero and Self-Calibration.

SERVICE

Conference Reviewer: ARR 2024-present, ICLR 2025-present, COLM 2024-present, AAAI 2026, CVPR 2026, ICML 2026.